
CONCEPT SUMMARY: n th Roots, Radicals, and Rational Exponents

Relating Radical and Exponential Forms

Radical Form

Exponential Form

Numbers

$$\sqrt[5]{32^4} = (32^4)^{\frac{1}{5}} = 32^{\frac{4}{5}}$$

$$729^{\frac{5}{6}} = (729^{\frac{1}{6}})^5 = (\sqrt[6]{729})^5$$

Algebra

$$\sqrt[n]{c^m} = (c^m)^{\frac{1}{n}} = c^{\frac{m}{n}}$$

$$c^{\frac{m}{n}} = (c^{\frac{1}{n}})^m = \sqrt[n]{c^m}$$
